

Original Contribution

Long-Term Exposure to Ambient Fine Particulate Matter and Mortality From Renal Failure: A Retrospective Cohort Study in Hong Kong, China

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Numerous studies have indicated that ambient particulate matter is closely associated with increased risk of cardiovascular disease, yet the evidence for its association with renal disease remains underrecognized. We aimed to estimate the association between long-term exposure to fine particulate matter, defined as particulate matter with an aerodynamic diameter less than or equal to 2.5 μm ($\text{PM}_{2.5}$), and mortality from renal failure (RF) among participants in the Elderly Health Service Cohort in Hong Kong, China, from 1998 to 2010. $\text{PM}_{2.5}$ concentration at the residential address of each participant was estimated based on a satellite-based spatiotemporal model. We used Cox proportional hazards regression to estimate risks of overall RF and cause-specific mortality associated with $\text{PM}_{2.5}$. After excluding 5,373 subjects without information on residential address or relevant covariates, we included 61,447 participants in data analyses. We identified 443 RF deaths during the 10 years of follow-up. For an interquartile-range increase in $\text{PM}_{2.5}$ concentration (3.22 $\mu\text{g}/\text{m}^3$), hazard ratios for RF mortality were 1.23 (95% confidence interval: 1.06, 1.43) among all cohort participants and 1.42 (95% confidence interval: 1.16, 1.74) among patients with chronic kidney disease. Long-term exposure to atmospheric $\text{PM}_{2.5}$ might be an important risk factor for RF mortality in the elderly, especially among persons with existing renal diseases.

air pollution; chronic kidney disease; fine particulate matter; kidney disease; particulate matter; renal failure; residential exposure

Abbreviations: AKI, acute kidney injury; BMI, body mass index; CI, confidence interval; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate; HR, hazard ratio; IQR, interquartile range; $\text{PM}_{2.5}$, particulate matter with an aerodynamic diameter less than or equal to 2.5 μm ; PM_{10} , particulate matter with an aerodynamic diameter less than or equal to 10 μm ; RF, renal failure; SEC, surface extinction coefficient.

The links between ambient air pollution and increased risk of cardiometabolic diseases such as hypertension and diabetes mellitus have been well investigated (1–5). Although the biological mechanisms regarding the associations warrant further exploration, mounting evidence indicates that the potential pathways of oxidative stress, systemic inflammation, and disturbance of vascular tone might be contributors (6–9). It is also hypothesized that impaired renal function may be related to cardiovascular impacts of air pollution, because the kidney is an organ that is also sensitive to oxidative stress, microvascular dysfunction, and systemic inflammation (10, 11). Although the roles of air pollution in the incidence and development of hypertension and diabetes

have been well documented in previous epidemiologic and clinical studies (10–14) and it is widely accepted that hypertension and diabetes are major risk factors for both acute kidney injury (AKI) and chronic kidney disease (CKD) (10, 15), direct human epidemiologic evidence regarding the relationship between air pollution and renal diseases is limited.

Fine particulate matter, defined as particulate matter with an aerodynamic diameter less than or equal to 2.5 μm ($\text{PM}_{2.5}$), is mainly a product of fossil fuel combustion from vehicles, power plants, etc. Some (but not all) epidemiologic studies have suggested that $\text{PM}_{2.5}$ may increase the risks of onset of CKD and end-stage renal disease

(16–22). In a cross-sectional US study conducted among stroke patients, Lue et al. (23) observed that residential proximity to major roadways was inversely related to estimated glomerular filtration rate (eGFR). Subsequent longitudinal studies indicated that chronic exposure to high PM_{2.5} density contributed to a decline in age-related eGFR in older men (17, 18). In another population-based study conducted in Taiwanese adults, Yang et al. (21) observed that reduced renal function was associated with particulate matter with an aerodynamic diameter less than or equal to 10 µm (PM₁₀) but not PM_{2.5}. Experimental studies have suggested that exposure to traffic emissions might promote renal oxidative stress and DNA damage, disturb renal hemodynamics, and exacerbate acute kidney injuries, but the findings are still controversial (24, 25). In a recent study, Yan et al. (26) also observed glomerulosclerosis in rats exposed to ambient particles over a period of 16 weeks.

Renal failure (RF) denotes the progressive loss of kidney function, including the subcategories AKI, CKD, and unspecified RF. In this study, we evaluated the associations between long-term PM_{2.5} exposure and risk of RF-related mortality among elderly participants in a longitudinal cohort study in Hong Kong, China (27). We used satellite-based data to measure yearly average PM_{2.5} exposure in each individual. The aim of our study was to explore the associations between chronic PM_{2.5} exposure and mortality from RF among all cohort participants and among those patients with existing CKD.

METHODS

Study population

A total of 66,820 participants were enrolled in the Hong Kong Department of Health Elderly Health Service Cohort from 1998 to 2000, which accounts for approximately 9% of the elderly population (persons aged ≥65 years) in Hong Kong. Enrollment was voluntary. The Elderly Health Centre located in each district of Hong Kong provided regular health assessments via structured and standardized interviews, as well as comprehensive clinical examinations. Physicians and registered nurses collected subjects' information, including data on sociodemographic characteristics, lifestyle, health behaviors, and other factors. Data on self-reported hypertension, diabetes, heart disease, stroke, and chronic obstructive pulmonary disease (COPD)/asthma, as individual covariates, were also collected by questionnaire at baseline. Details were reported in an earlier paper (27). Health assessments were administered during both the baseline period (1998–2000) and the follow-up period (2001–2010). Approximately 70% of participants were still enrolled within 3 years after baseline assessments, indicating high follow-up compliance. Information on morbidity and mortality was extracted through record linkages with the Hong Kong Hospital Authority and the Hong Kong Census and Statistics Department, respectively. Approval for the research protocol was obtained from the Hospital Authority Hong Kong West Cluster and the Institutional Review Board of the University of Hong Kong.

Outcome ascertainment

The causes of morbidity and mortality were encoded according to the Ninth and Tenth Revisions of the *International Classification of Diseases*. Morbidity records from 1998–2010 and mortality records from 1998–2001 were coded using the *International Classification of Diseases, Ninth Revision* (RF: codes 584.5–586; AKI: codes 584.5–854.9; CKD: codes 585.1–585.9; unspecified RF: code 586). Mortality records obtained after 2001 were coded using the *International Classification of Diseases, Tenth Revision* (RF: codes N17–N19; AKI: code N17; CKD: code N18; unspecified RF: code N19). The primary diagnosis associated with the death was used as the cause of death. Almost all death events in Hong Kong occurred in hospitals with accurate ascertainment of cause of death (27). Subjects who died during the first follow-up year after baseline were excluded from the analysis.

Exposure estimation model

PM_{2.5} exposure measurements were deduced from satellite remote sensing data which were calibrated with ground meteorological observations (1, 28). In brief, aerosol optical depth data obtained from 2 Earth Observing System satellite sensors (US National Aeronautics and Space Administration) were treated as indicators for the level of particulate matter in ambient air. Atmospheric PM_{2.5} was measured by calculating surface extinction coefficients (SECs) according to aerosol optical depth at a 1 km × 1 km spatial resolution after accounting for rain-related humidity (28, 29). Surface observations of hourly PM_{2.5} concentrations from 1998 to 2010 were obtained from monitoring stations governed by the Hong Kong Environmental Protection Department. The model of PM_{2.5} exposure was further derived by averaging ground-level air concentrations on a yearly basis and then regressing the values over the corresponding annual average satellite SEC data. Cross-validation tests were used to confirm the validity of the approach. In short, we used concentrations from 3 sites for building the regression model (PM_{2.5} approximated SEC) and then used the model to predict concentrations at the fourth site from 2000 to 2011. We found that predicted annual PM_{2.5} differed from the observed data obtained from the Environmental Protection Department by approximately 9%–12% (1, 28, 29). Annual PM_{2.5} assessments were made with the exposure models between 1998 and 2010 and then matched with individual participants by geocoded residential location as the explanatory variable. In this elderly Chinese cohort, approximately 13.3% of participants had a change in residential address during the study period, which was taken into account when estimating annual exposure (29).

Individual, ecological, and environmental covariates

We included individual, ecological, and environmental covariates as potential confounders in the regression models of air pollution and health (19, 21). Individual-level covariates included age, sex, body mass index (BMI; weight (kg)/height (m)²), alcohol consumption, cigarette smoking,

education, monthly household expenditures, and frequency of physical exercise (days/week). Self-reported data on comorbid conditions (hypertension, diabetes, heart disease, stroke, and COPD/asthma) as individual covariates were also collected by questionnaire at baseline. Access to health care for participants was relatively equitable because of the virtually free public health care in Hong Kong, which acts as a safety net for the whole community (30). We did not adjust for population density as a potential confounder because it was not found to be a predictor of RF mortality on the basis of district-level data (see Web Figure 1, available at <https://academic.oup.com/aje>). Ecological covariates pertained to neighborhood characteristics and were based on 197 Hong Kong tertiary planning units, including the percentage of elderly persons (ages ≥ 65 years), the percentage of persons with a tertiary educational level (secondary school or above), and the percentage of households with an income greater than or equal to US\$1,923/month (31). For estimation of duration (years) of secondhand smoke exposure at baseline, the percentage of smokers in the district was included as an environmental covariate (31).

Statistical analysis

We utilized Cox proportional hazards regression models to fit the relationship between annual $PM_{2.5}$ exposure and RF mortality among all cohort participants and among patients with existing CKD. We also estimated the associations between particulate matter exposure and the development of incident renal diseases, including AKI, CKD, and unspecified RF. Attained age was chosen as the underlying time scale, since it automatically controlled for age as a confounding factor (32). Yearly $PM_{2.5}$ concentration on the city level was included in the model as a time-varying exposure of interest. The Cox proportional hazards assumption was tested before conducting regression analysis.

We estimated hazard ratios per interquartile-range (IQR) increment in $PM_{2.5}$ level in regression models with hierar-

chical strategies for covariate adjustment (1, 29). With age adopted as the underlying time-scale variable, we controlled for calendar year of entry and sex in model 1. Besides the covariates included in model 1, we additionally controlled for more individual covariates in model 2, including BMI, alcohol consumption, cigarette smoking, education, physical activity level, medication use, monthly household expenditures, and self-reported comorbidity at baseline (hypertension, diabetes, heart disease, stroke, and COPD/asthma). We further controlled for ecological and environmental covariates in model 3, including tertiary planning unit characteristics (percentage of elderly persons, percentage with a tertiary education, and percentage with a per capita household income of at least US\$1,923 per month) and percentage of smokers in the district.

We performed sensitivity analyses by: 1) not excluding subjects who died during the first year of follow-up after baseline; 2) excluding the deaths with less than 3 years of follow-up since enrollment; and 3) repeating the analysis on the association of interest among patients with existing RF and all renal diseases only. We performed several stratification analyses to examine potential heterogeneity by sex, BMI, and self-reported hypertension, diabetes, and heart disease at baseline. We examined the statistical significance of differences across subgroups by means of the algorithm $(\beta_1 - \beta_2) \pm 1.96 \sqrt{SE_1^2 + SE_2^2}$, where β_1 and β_2 are estimates for each subgroup and SE_1 and SE_2 are the corresponding standard errors, respectively (33). The data were analyzed using R statistical software, version 3.3.2 (R Foundation for Statistical Computing, Vienna, Austria), and the R “survival” package.

RESULTS

A total of 66,820 participants were initially enrolled in the program. We excluded 1,932 participants (2.9%) for having missing residential addresses for geocoding, 3,420

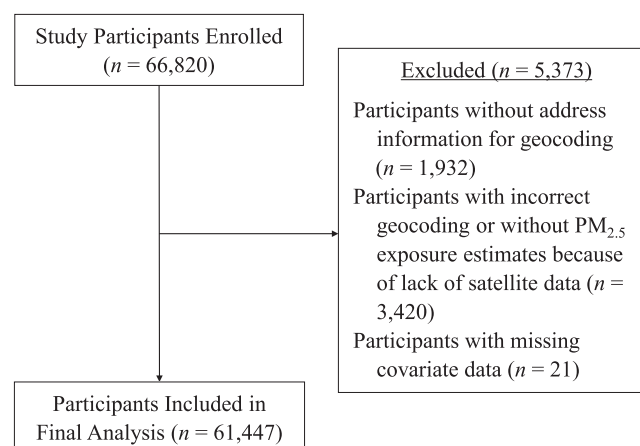


Figure 1. Process used to select cohort participants for an analysis of the association between long-term exposure to fine particulate matter, defined as particulate matter with an aerodynamic diameter less than or equal to $2.5 \mu m$ ($PM_{2.5}$), and mortality from renal failure, Hong Kong, China, 1998–2010.

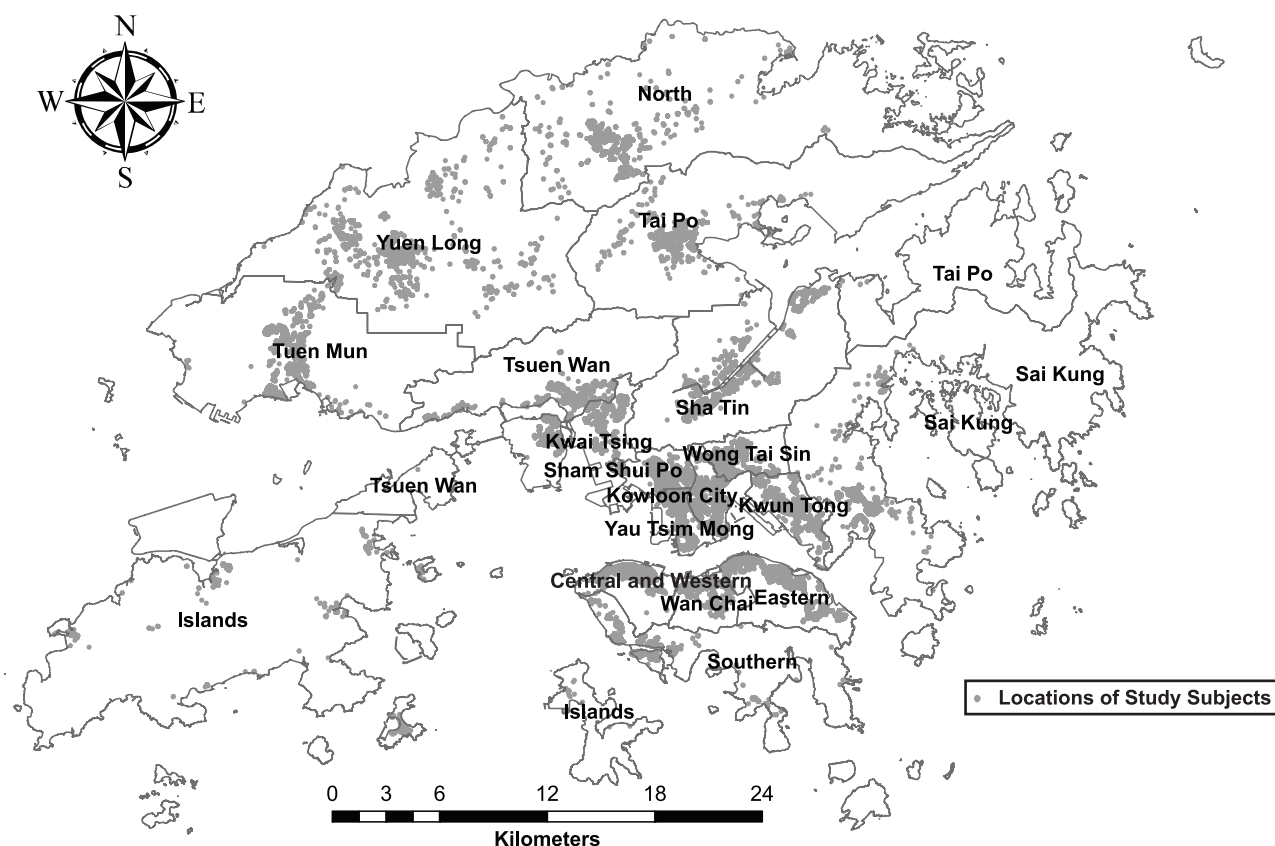


Figure 2. Spatial distribution of participants ($n = 61,447$) in a cohort of elderly persons (ages ≥ 65 years) at baseline, Hong Kong, China, 1998–2000.

participants (5.1%) for having missing $PM_{2.5}$ exposure estimates or incorrect geocoding, and 21 participants (0.03%) for having missing covariate data (Figure 1). The spatial distribution of the remaining 61,447 subjects with complete data is shown in Figure 2. The mean age of participants at cohort entry was 72.0 years, and the proportion of women was 65.9%. Approximately 7.1% of participants were categorized as underweight, while 40.2% were classified as overweight or obese. With regard to lifestyle, 28.9% were former or current smokers, 13.8% were former or regular alcohol drinkers, and 15.3% never exercised. The majority of participants (82.9%) had a primary school education or less. Over half of the participants (53.1%) regularly took medication. With regard to self-reported comorbidity, 36.0% of enrolled participants had hypertension, 12.3% had diabetes, 12.4% had heart disease, 3.0% had had a stroke, and 5.6% had COPD or asthma (Table 1). No participants reported having any kidney diseases on the baseline questionnaire, and identification of incident CKD was based on data linkage to hospital admission records during the follow-up period.

During the baseline period from 1998 to 2000, the median $PM_{2.5}$ concentration was $35.78 \mu g/m^3$, and the corresponding IQR was $3.22 \mu g/m^3$. The median value was chosen as the cutoff point for differentiating between high and

low exposure categories. Table 1 shows the distribution of covariates according to level of particulate matter exposure. Individual exposure to $PM_{2.5}$ among the cohort participants followed a normal distribution (Web Figure 1). The correlation between annual $PM_{2.5}$ concentration obtained from satellite-based SECs and that obtained from general monitoring stations at baseline was 0.687 ($P < 0.001$).

In the Elderly Health Service Cohort, 443 RF deaths were identified during 10 years of follow-up. In all 3 models, elevated $PM_{2.5}$ exposure was statistically significantly associated with a higher risk of RF mortality among all cohort participants and among incident CKD patients. The fully adjusted (model 3) hazard ratio for RF mortality per IQR increment in $PM_{2.5}$ ($3.22 \mu g/m^3$) among all cohort participants was 1.23 (95% confidence interval (CI): 1.06, 1.43) (Table 2). During 10 years of follow-up, we identified 1,204 patients with incident CKD, and the hazard ratio of interest increased to 1.42 (95% CI: 1.16, 1.74) in the subpopulation of CKD patients (Table 2). In further subgroup analyses, an IQR increment in $PM_{2.5}$ concentration led to an elevated risk of AKI mortality (hazard ratio (HR) = 2.03, 95% CI: 1.34, 3.08) but not mortality from CKD (HR = 1.14, 95% CI: 0.96, 1.36) or unspecified RF (HR = 1.19, 95% CI: 0.80, 1.78) (Table 3). For the relationship between air pollution and the development of all incident renal diseases, the risk estimates

Table 1. Baseline Characteristics of Participants in the Elderly Health Service Cohort ($n = 61,447$) by $PM_{2.5}$ Exposure, Hong Kong, China, 1998–2000

Variable	All Participants ($n = 61,447$)		$PM_{2.5}$ Exposure Level ^a			
			Low ($<35.8 \mu g/cm^3$) ($n = 30,716$)		High ($\geq 35.8 \mu g/cm^3$) ($n = 30,731$)	
	No.	%	No.	%	No.	%
Age at cohort entry, years ^b	72.0 (5.6)		72.0 (5.6)		72.1 (5.6)	
Sex						
Male	20,933	34.1	10,539	34.3	10,394	33.8
Female	40,514	65.9	20,177	65.7	20,337	66.2
Body mass index ^c category						
Underweight (<18.5)	4,368	7.1	2,106	6.9	2,262	7.4
Normal weight (18.5–24.9)	32,370	52.7	15,978	52.0	16,392	53.3
Overweight (25.0–29.9)	20,921	34.0	10,588	34.5	10,333	33.6
Obese (≥ 30.0)	3,788	6.2	2,044	6.7	1,744	5.7
Cigarette smoking						
Never smoker	43,669	71.1	21,947	71.5	21,722	70.7
Former smoker	11,871	19.3	5,881	19.1	5,990	19.5
Current smoker	5,907	9.6	2,888	9.4	3,019	9.8
Alcohol consumption						
Never/social drinker	52,986	86.2	26,497	86.3	26,489	86.2
Former/regular drinker	8,461	13.8	4,219	13.7	4,242	13.8
Physical exercise, days/week						
None (0)	9,406	15.3	4,314	14.0	5,092	16.6
Medium (1–6)	7,789	12.7	3,839	12.5	3,950	12.9
High (7)	44,252	72.0	22,563	73.5	21,689	70.6
Educational level						
Below primary school	28,242	46.0	14,415	46.9	13,827	45.0
Primary school	22,656	36.9	10,931	35.6	11,725	38.2
Secondary school or above	10,549	17.2	5,370	17.5	5,179	16.9
Monthly household expenses						
Low ($<US\$128$)	10,122	16.5	4,323	14.1	5,799	18.9
Medium ($US\$128$ – $US\$384$)	42,152	68.6	21,474	69.9	20,678	67.3
High ($\geq US\$385$)	9,173	14.9	4,919	16.0	4,254	13.8
Medication use						
Yes	32,628	53.1	16,262	52.9	16,366	53.3
No	28,819	46.9	14,454	47.1	14,365	46.7
Hypertension						
Yes	22,123	36.0	10,987	35.8	11,136	36.2
No	39,324	64.0	19,729	64.2	19,595	63.8
Diabetes mellitus						
Yes	7,542	12.3	3,742	12.2	3,800	12.4
No	53,905	87.7	26,974	87.8	26,931	87.6
Heart disease						
Yes	7,618	12.4	3,648	11.9	3,970	12.9
No	53,829	87.6	27,068	88.1	26,761	87.1

Table continues

Table 1. Continued

Variable	All Participants (n = 61,447)		PM _{2.5} Exposure Level ^a			
			Low (<35.8 µg/cm ³) (n = 30,716)		High (≥35.8 µg/cm ³) (n = 30,731)	
	No.	%	No.	%	No.	%
Stroke						
Yes	1817	3.0	831	2.7	986	3.2
No	59,630	97.0	29,885	97.3	29,745	96.8
Chronic obstructive pulmonary disease						
Yes	3,458	5.6	1,702	5.5	1,756	5.7
No	57,989	94.4	29,014	94.5	28,975	94.3
TPU-level covariates ^b						
% of persons aged ≥65 years	12.11 (4.19)		11.39 (4.08)		12.83 (4.17)	
% of persons with a tertiary education	12.98 (7.96)		13.33 (8.24)		12.64 (7.66)	
% of households with income ≥US\$1,923/month	59.50 (11.60)		61.87 (10.95)		57.13 (11.75)	

Abbreviations: PM_{2.5}, particulate matter with an aerodynamic diameter less than or equal to 2.5 µm; TPU, tertiary planning unit.

^a Low and high PM_{2.5} exposure were defined using the median value (35.7756 µg/cm³) as the cutoff point.

^b Values are presented as mean (standard deviation).

^c Weight (kg)/height (m)².

for AKI and CKD per IQR increase in PM_{2.5} level were 1.20 (95% CI: 1.04, 1.38) and 0.95 (95% CI: 0.87, 1.03), respectively (Web Table 1).

The overall associations between fine particulate matter and RF mortality risk among all participants and CKD patients were close to linear (Figure 3). There was some indication that dose-response curves leveled off when PM_{2.5} concentrations exceed 40 µg/m³. In subgroup analyses, we did not find risk estimate differences by sex, BMI, self-reported hypertension, diabetes, or heart disease (Table 4). Consistent results were generated in the series of sensitivity

analyses including participants who died during the first year of follow-up, excluding deaths with less than 3 years of follow-up, and estimating the association between PM_{2.5} and RF-related mortality among patients with existing RF and all renal diseases (Web Tables 2 and 3).

DISCUSSION

In this study of an elderly cohort in Hong Kong, particulate matter air pollution was associated with elevated risks of RF mortality among all cohort subjects and especially

Table 2. Hazard Ratio for Renal Failure Mortality per Interquartile-Range Increase in Ambient PM_{2.5} Concentration in the Elderly Health Service Cohort, Hong Kong, China, 1998–2010

Model	All Participants				CKD Patients			
	No. of Deaths	HR	95% CI	P Value	No. of Deaths	HR	95% CI	P Value
Model 1 ^a	443	1.24	1.08, 1.43	0.002	253	1.56	1.30, 1.87	<0.001
Model 2 ^b	443	1.27	1.11, 1.46	0.001	253	1.57	1.31, 1.90	<0.001
Model 3 ^c	443	1.23	1.06, 1.43	0.007	253	1.42	1.16, 1.74	0.001

Abbreviations: CI, confidence interval; CKD, chronic kidney disease; HR, hazard ratio; PM_{2.5}, particulate matter with an aerodynamic diameter less than or equal to 2.5 µm; TPU, tertiary planning unit.

^a Model 1 controlled for calendar year of cohort entry and sex.

^b Model 2 controlled for baseline levels of individual covariates, sex, body mass index, calendar year of cohort entry, cigarette smoking, education, alcohol consumption, monthly household expenses, physical activity level, medication use, hypertension, diabetes mellitus, heart disease, stroke, and chronic obstructive pulmonary disease/asthma.

^c Model 3 controlled for TPU-level covariates (percentage of persons aged ≥65 years, percentage of persons with a tertiary education, and percentage of households with income of ≥US\$1,923/month), district-level smoking prevalence, and all covariates in model 2.

Table 3. Hazard Ratios for Specific Types of Renal Failure Mortality per Interquartile-Range Increase in Ambient PM_{2.5} Concentration in the Elderly Health Service Cohort, Hong Kong, China, 1998–2010

Model	AKI			CKD			Unspecified RF		
	No. of Deaths	HR	95% CI	No. of Deaths	HR	95% CI	No. of Deaths	HR	95% CI
Model 1 ^a	63	1.83	1.27, 2.65	319	1.16	0.99, 1.36	58	1.34	0.92, 1.97
Model 2 ^b	63	1.82	1.26, 2.65	319	1.18	1.01, 1.39	58	1.39	0.95, 2.05
Model 3 ^c	63	2.03	1.34, 3.08	319	1.14	0.96, 1.36	58	1.19	0.80, 1.78

Abbreviations: AKI, acute kidney injury; CI, confidence interval; CKD, chronic kidney disease; HR, hazard ratio; PM_{2.5}, particulate matter with an aerodynamic diameter less than or equal to 2.5 µm; RF, renal failure; TPU, tertiary planning unit.

^a Model 1 controlled for calendar year of cohort entry and sex.

^b Model 2 controlled for baseline levels of individual covariates, sex, body mass index, calendar year of cohort entry, cigarette smoking, education, alcohol consumption, monthly household expenses, physical activity level, medication use, hypertension, diabetes mellitus, heart disease, stroke, and chronic obstructive pulmonary disease/asthma.

^c Model 3 controlled for TPU-level covariates (percentage of persons aged ≥65 years, percentage of persons with a tertiary education, and percentage of households with income of ≥US\$1,923/month), district-level smoking prevalence, and all covariates in model 2.

among CKD patients. In a more detailed analysis of the entire natural history of kidney disease, we did not find an association between air pollution and the development of incident CKD. Therefore, it is likely that the overall association between air pollution and RF mortality reflects mainly the progression of CKD exacerbated by air pollution. However, caution is needed for this interpretation, because the identification of incident CKD cases was based on self-reports and record linkage to public hospital admission data. No participants reported having kidney disease on the baseline questionnaire, but it was possible that a fraction of the cohort participants were already CKD patients; on the other hand, persons admitted to a hospital because of CKD might represent just a fraction of the incident CKD cases whose symptoms were severe enough for hospitalization. In comparison with the identification of incident CKD cases, the renal disease mortality data should have been more accurate, because almost all deaths in Hong Kong occurred in hospitals with accurate and consistent death ascertainment (27). Therefore, the current findings contribute to the evidence base on the relationship between long-term air pollution exposure and RF mortality in the elderly population.

Our findings were generally consistent with previous studies on air pollution and renal diseases (16–22, 34). While the estimated effect sizes in those studies varied among the populations of the United States, mainland China, Taiwan, South Korea, and Hong Kong, air pollution was consistently associated with elevated risks of kidney disease (16–22, 34). In the current study, we found relatively higher risks of RF mortality associated with PM_{2.5}, especially among CKD patients, in this elderly Hong Kong cohort, but in a previous study the CKD incidence associated with PM_{2.5} was higher in a US population than in an Asian population (16–19). Specifically, cross-sectional studies carried out in the United States found an association between PM_{2.5} and decreases in eGFR and CKD prevalence (16, 19). Studies in a cohort of US veterans found that a 10-µg/m³ increase in PM_{2.5} and PM₁₀ concentrations corresponded

to 27% and 7% increases in CKD incidence, respectively (17, 18). Investigators in a series of cross-sectional studies also found that exposure to PM₁₀ or PM_{2.5} was associated with increased risks of membranous nephropathy, eGFR decline, and CKD prevalence in an Asian population (21, 22, 34); and in a cohort study in Taiwan, it was estimated that a 10-µg/m³ increment in PM_{2.5} concentration corresponded to a 6% increase in CKD incidence (20).

Fine particulate matter air pollution was associated with elevated AKI mortality in the current study. AKI is a syndrome characterized by rapid loss of the kidney's excretory function and is interconnected with CKD (35). Risk factors for AKI include diabetes, cardiovascular disease, cancer, and complex surgery (15). The association between air pollution and AKI might relate to PM_{2.5} inhalation-derived systemic inflammation and the subsequent exacerbation of AKI symptoms (8, 36). In stratification analysis, we found weak differences in the association between PM_{2.5} concentration and RF mortality by sex and BMI. The findings in this Hong Kong elderly population were consistent with observations from the Taipei, Taiwan, metropolitan area (21). We did not observe risk estimate differences between participants with and without self-reported hypertension, diabetes, or heart disease at baseline. Potential heterogeneity in the relationship between air pollution and renal disease by comorbidity should be examined more systematically in future studies.

The possible biological mechanisms underlying the impact of air pollutants on the renal system remain largely uncertain. Mehta et al. (19) hypothesized that the adverse association was mediated by an increase in diastolic blood pressure. Numerous studies have indicated an association between PM_{2.5} exposure and markers of endothelial/vascular dysfunction, which might result in serum creatinine-derived eGFR decline, CKD, and end-stage renal disease (37, 38). However, few epidemiologic studies have explored the mediation function of diastolic blood pressure in the pathway from ambient air pollution to renal disease. A second hypothesis relates to oxidative stress and air pollution-induced inflammatory mediators, including interleukin 6,

Table 4. Stratified Analyses of the Association (Model 3) Between PM_{2.5} Exposure and Renal Failure Mortality, Overall and by Subtype, in the Elderly Health Service Cohort, Hong Kong, China, 1998–2010^a

Characteristic	Overall RF			AKI			CKD			Unspecified RF		
	HR	95% CI	P Value ^b	HR	95% CI	P Value	HR	95% CI	P Value	HR	95% CI	P Value
Sex												
Male	1.18	0.93, 1.49		3.58	1.56, 8.25		1.17	0.89, 1.52		0.75	0.42, 1.34	
Female	1.27	1.04, 1.54	0.638	1.75	1.08, 2.86	0.147	1.13	0.90, 1.43	0.865	1.90	1.07, 3.40	0.026
BMI ^c category												
Normal weight (18.5–24.9)	1.35	1.09, 1.67		2.22	1.21, 4.06		1.32	1.03, 1.70		1.08	0.63, 1.85	
Underweight (<18.5)	1.08	0.79, 1.47	0.242	1.71	0.82, 3.60	0.599	0.91	0.62, 1.32	0.099	1.51	0.65, 3.53	0.516
Overweight/obese (≥25)	1.16	0.86, 1.55	0.760	2.30	0.92, 5.75	0.627	1.05	0.75, 1.47	0.558	1.41	0.60, 3.29	0.910
Hypertension												
No	1.14	0.88, 1.48		2.01	1.09, 3.68		0.99	0.71, 1.37		1.21	0.67, 2.17	
Yes	1.28	1.07, 1.54	0.483	2.08	1.18, 3.68	0.928	1.22	0.99, 1.50	0.282	1.25	0.72, 2.17	0.932
Diabetes mellitus												
No	1.20	1.00, 1.44		2.05	1.29, 3.26		1.07	0.86, 1.33		1.33	0.83, 2.12	
Yes	1.30	1.00, 1.70	0.630	1.90	0.75, 4.84	0.890	1.29	0.96, 1.74	0.330	0.98	0.45, 2.10	0.501
Heart disease												
No	1.28	1.08, 1.52		2.13	1.35, 3.34		1.19	0.97, 1.45		1.26	0.79, 1.99	
Yes	1.05	0.76, 1.47	0.301	1.63	0.57, 4.70	0.652	1.02	0.69, 1.49	0.483	1.02	0.42, 2.51	0.689

Abbreviations: AKI, acute kidney injury; BMI, body mass index; CI, confidence interval; CKD, chronic kidney disease; HR, hazard ratio; PM_{2.5}, particulate matter with an aerodynamic diameter less than or equal to 2.5 µm; RF, renal failure; TPU, tertiary planning unit.

^a Model 3 controlled for all individual, ecological, and environmental covariates, including sex, BMI, calendar year of cohort entry, cigarette smoking, education, alcohol consumption, monthly household expenses, physical activity level, medication use, hypertension, diabetes mellitus, heart disease, stroke, and chronic obstructive pulmonary disease/asthma, as well as TPU-level covariates (percentage of persons aged ≥65 years, percentage of persons with a tertiary education, and percentage of households with income of ≥US\$1,923/month) and district-level smoking prevalence.

^b Wald test for the difference in HRs between categories, such as male versus female sex (for BMI, comparisons were for normal weight versus underweight and normal weight versus overweight/obese).

^c Weight (kg)/height (m)².

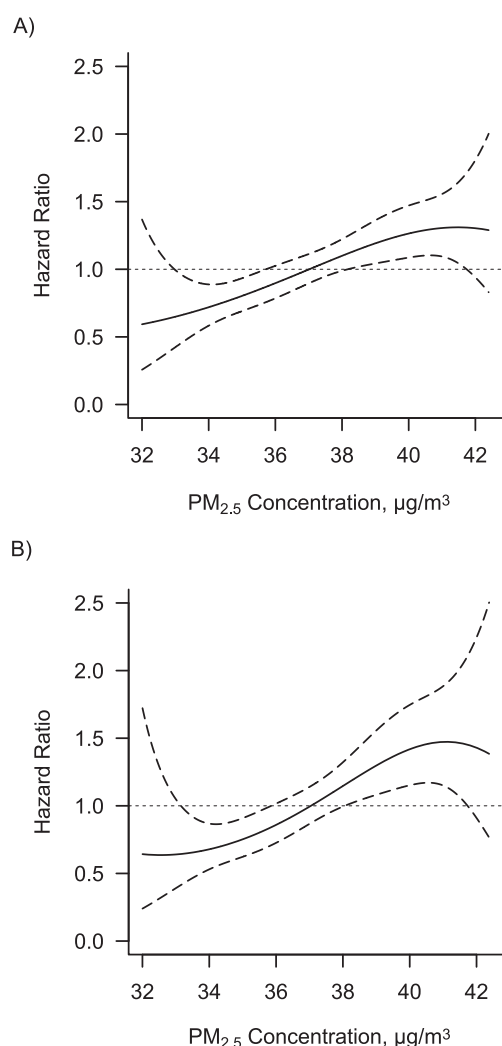


Figure 3. Exposure-response relationships between fine particulate matter and mortality from renal failure among all participants (A) and patients with chronic kidney disease (B) in the Elderly Health Service Cohort, Hong Kong, China, 1998–2010. $PM_{2.5}$, particulate matter with an aerodynamic diameter less than or equal to 2.5 μm . Dashed lines, 95% confidence intervals.

tumor necrosis factor α , and plasminogen activator inhibitor 1. These immunological molecules may exaggerate vasoconstrictor responses to serotonin and phenylephrine, resulting in severe renal disease (6–8, 39). A third hypothesis suggests that ambient air pollution might result in metabolic disturbances, including decreased insulin sensitivity, glucose intolerance, weight gain, a higher blood lipid concentration, and diabetes, most of which are risk factors for AKI, CKD, and end-stage renal disease (40, 41). It is likely that more than 1 mechanistic pathway is involved in the association between particulate matter air pollution and various renal diseases.

A few limitations of the present study need to be noted. First, participants were enrolled on a voluntary basis, and thus they might have been more health-conscious than the general elderly population in Hong Kong. Indeed, there were

fewer smokers and alcohol drinkers in the cohort than in the general Hong Kong population (Table 1). Moreover, there were more female participants (65.9%) in this cohort than in the overall elderly population of Hong Kong. Caution is needed to interpret the generalizability of the current findings. Second, we used SECs from aerosol optical depth within 1 km $\times$ 1 km of ground level to predict air concentrations of particulate matter. Exposure variability was relatively modest for $PM_{2.5}$ (the IQR was approximately equal to 9% of the mean) (28), which was a restraint on statistical power in the current study. Third, we did not perform specific clinical checks on the participants' renal function at baseline, especially their eGFR. There might have been some enrolled participants who had slight renal dysfunction but no obvious clinical symptoms. Fourth, residual confounding was still possible. Socioeconomic status was only controlled for by educational level, monthly household expenditures, and some district-level covariates, and information on the use of hypertensive or diabetic medication was not available for the current analysis.

In conclusion, long-term exposure to fine particulate matter air pollution might be an important risk factor for RF mortality in the elderly, especially among persons with existing CKD. These population-level findings contribute to the evidence base on the relationship between air pollution and renal diseases and should be further complemented by experimental and clinical studies.

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